

Cyclic Groups

A group is **cyclic** if all the elements of the group can be generated by repeatedly applying the group operation to a single element of the group. This element is called a **generator** of the group. Let's look at an example of a cyclic group.

Example: We already know that $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$ is a group under multiplication modulo 5. It also happens to be cyclic because all of its elements can be generated by the element 2 (in fact, 3 can also generate the group). Here's what I mean:

$$\begin{aligned} 2 &= 2 \\ 2 \times 2 \pmod{5} &= 4 \pmod{5} = 4 \\ 2 \times 2 \times 2 \pmod{5} &= 8 \pmod{5} = 3 \\ 2 \times 2 \times 2 \times 2 \pmod{5} &= 16 \pmod{5} = 1. \end{aligned}$$

We can generate every element in $\{1, 2, 3, 4\}$ by multiplying 2 by itself $\pmod{5}$ over and over again. Because of this, we can write

$$(\mathbb{Z}/5\mathbb{Z})^\times = \langle 2 \rangle.$$

This notation tells us that $(\mathbb{Z}/5\mathbb{Z})^\times$ is a cyclic group generated by the element 2.

Exercises

(1.) Verify that $(\mathbb{Z}/5\mathbb{Z})^\times = \langle 3 \rangle$, i.e., that 3 is also a generator of $(\mathbb{Z}/5\mathbb{Z})^\times$.

(2.) Interestingly, $(\mathbb{Z}/p\mathbb{Z})^\times$ is always cyclic if p is a prime number. Thus, you can find a generator for $(\mathbb{Z}/7\mathbb{Z})^\times$. What is it? There might be more than one.